Lexical and Semantic Search for French PhD Theses

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Abstract

This project studies information retrieval for French PhD theses collected from theses.fr. The goal is to compare a classical lexical retrieval approach based on TF-IDF and cosine similarity with a semantic retrieval approach based on transformer sentence embeddings. The corpus contains 18,242 unique theses after deduplication. Since abstracts are not directly returned by the theses.fr search endpoint, an additional enrichment step extracts available abstracts from individual thesis pages. This step provides 16,831 non-empty abstracts, corresponding to an abstract coverage of 92.27%. The final searchable representation concatenates the main title, the English title when available, and the abstract. Other fields such as discipline, subjects, institution, year, status and URL are kept as metadata for exploratory analysis, display and possible future filtering. This report presents the data collection pipeline, exploratory data analysis, the TF-IDF baseline, the semantic embedding retrieval method, and their manual comparison.

1 Introduction

Searching scientific documents is a natural information retrieval problem: given a user query, the system must return documents that are relevant to the user's information need. In this project, the documents are French PhD theses collected from theses.fr, and the objective is to build a search engine able to rank theses according to a natural language query.

A simple keyword-based search may work when the query and the document share exact terms. However, it can fail when the same idea is expressed with different words. For instance, a query about "deep learning for medical imaging" may be relevant to theses mentioning "neural networks", "radiology", or "computer-aided diagnosis", even if the exact query terms do not appear in the title.

The project compares two families of retrieval methods. The first one is a classical lexical baseline based on TF-IDF vectors and cosine similarity. The second one is a semantic approach based on transformer sentence embeddings. The main question is whether dense semantic embeddings can retrieve more relevant theses than a sparse lexical baseline on French academic metadata and abstracts.

The contribution of this work is threefold:

- building a reproducible pipeline to collect and preprocess theses.fr data;
- implementing lexical and semantic retrieval models on the same corpus;
- comparing their results through qualitative queries and simple quantitative ranking analyses.

Preprint.

2 Dataset and data collection

2.1 Data source

The dataset is collected from theses.fr, the French search engine for doctoral theses. The search API returns structured metadata for theses, including identifiers, titles, disciplines, subjects, institutions, status and dates. The collection step uses the endpoint:

`https://theses.fr/api/v1/theses/recherche/`

The API is queried with several broad keywords related to artificial intelligence, machine learning, data science, computer science and statistics. The current collection was obtained with queries such as:

`intelligence artificielle, apprentissage automatique, deep learning, traitement automatique des langues, vision par ordinateur, science des données, modélisation statistique, réseaux de neurones, fouille de données, informatique`

After deduplication by thesis identifier, the raw dataset contains 18,242 theses.

2.2 Collected fields

The first collection step retrieves metadata from the search endpoint. Table 1 summarizes the main fields kept in the dataset.

Table 1: Main fields collected from theses.fr. The final retrieval models use a concatenation of the available textual fields.

Field	Description
`id`	Thesis identifier on theses.fr
`title`	Main thesis title
`title_en`	English title when available
`abstract`	Abstract extracted from the thesis page when available
`year`	Defense year or first registration year
`discipline`	Academic discipline
`subjects`	Keywords and topics returned by theses.fr
`institution`	Institution where the thesis is registered or defended
`status`	Thesis status, for example defended or ongoing
`url`	Link to the theses.fr page

2.3 Abstract enrichment

The theses.fr search endpoint does not directly return abstracts. However, abstracts are visible on individual thesis pages when available. The pipeline therefore includes a separate enrichment step that visits each thesis page and extracts the visible abstract from the HTML.

This step is separated from metadata collection for practical reasons. The search API is faster and more stable for retrieving structured metadata, whereas abstract extraction requires one HTTP request per thesis page and depends on the page structure. Keeping both steps separate makes the pipeline easier to debug and reproduce.

The enrichment was first validated on a subset of 1,000 theses and was then applied to the complete corpus. After enrichment, 16,831 out of 18,242 theses contain a non-empty abstract, corresponding to an abstract coverage of 92.27%. This high coverage makes the corpus suitable for comparing lexical and semantic retrieval methods. Nevertheless, the task should not be framed as search over abstracts only. The system performs retrieval over enriched theses.fr metadata, optionally completed with abstracts when available.

2.4 Preprocessing

The preprocessing step builds a final searchable field called `text`. This field is obtained by concatenating the main title, the English title when available, and the abstract:

```
text = title + title_en + abstract
```

Other fields such as `discipline`, `subjects`, `institution`, `year`, `status` and `url` are kept as metadata for exploratory analysis, result display and possible future filtering, but they are not included in the `text` field used in the reported retrieval experiments.

This design keeps the searchable representation simple and consistent across both retrieval methods. If an abstract is unavailable, the document can still be represented through its title and English title when available, although with less textual information.

The preprocessing remains intentionally light. It removes line breaks and repeated spaces, but it does not apply aggressive normalization. This choice is motivated by the fact that TF-IDF already includes its own tokenization, and transformer embeddings generally perform better on natural text.

We did not remove stopwords manually because TF-IDF already downweights very frequent terms through the inverse document frequency component, and aggressive stopword removal could remove useful academic expressions in French.

3 Exploratory data analysis

The exploratory data analysis aims to understand the size, completeness and structure of the collected corpus before training retrieval models. This step is important because retrieval quality depends not only on the model, but also on the amount and quality of textual information available for each document.

3.1 Corpus size and fields

After deduplication, the processed dataset contains 18,242 theses and 11 columns. The columns are: `id`, `title`, `title_en`, `abstract`, `year`, `discipline`, `subjects`, `institution`, `status`, `url`, and `text`.

The `text` column is added during preprocessing and is the field used by the retrieval models. It concatenates the main title, the English title when available, and the abstract.

3.2 Missing values

Table 2 reports the main missing values in the processed dataset. The most missing field is `subjects`, with 4,027 missing values, corresponding to 22.08% of the corpus. The `abstract` field has 1,411 missing values, corresponding to 7.73%. The English title is missing for 1,395 theses, or 7.65%. Only one thesis has a missing year, and the core fields used for identification and retrieval, such as `id`, `title`, `discipline`, `institution`, `status`, `url` and `text`, have no missing values.

The abstract coverage is therefore 92.27%, since 16,831 theses have a non-empty abstract. This is a useful property for semantic retrieval, because abstracts contain richer natural language descriptions than titles alone. However, 1,411 theses still have no abstract, which means that these documents are represented mainly through their title and English title when available. This limitation is important when interpreting retrieval results.

3.3 Text length

Table 3 summarizes the length of the main textual fields. The final `text` field has an average length of 2,069 characters and a median length of 1,910 characters. This confirms that most documents contain enough textual material for retrieval. However, the distribution is heterogeneous: some theses have very short descriptions, while the longest document representation contains more than 50,000 characters.

Table 2: Missing values in the processed dataset.

Field	Missing count	Missing rate (%)
`subjects`	4,027	22.08
`abstract`	1,411	7.73
`title_en`	1,395	7.65
`year`	1	0.01
`id`	0	0.00
`title`	0	0.00
`discipline`	0	0.00
`institution`	0	0.00
`status`	0	0.00
`url`	0	0.00
`text`	0	0.00

Table 3: Character length statistics for the main textual fields.

Statistic	Title length	Abstract length	Text length
Mean	103.80	1,877.16	2,068.83
Std. dev.	39.27	1,330.67	1,341.14
Min	7	0	34
25%	75	1,166.25	1,359
Median	99	1,697	1,910
75%	126	2,472	2,664
Max	468	50,132	50,460

This heterogeneity matters for lexical retrieval. A longer document has more opportunities to share exact words with a query, which may increase its TF-IDF similarity. This effect must be considered when interpreting search results.

3.4 Year distribution

The corpus contains both recent and older theses, with a strong concentration in recent years. Table 4 reports the number of theses for the most recent years in the dataset. The number of collected theses increases sharply after 2020, with 1,952 theses in 2023, 2,268 in 2024 and 2,236 in 2025. The year 2026 contains fewer theses because it is incomplete at collection time.

Table 4: Number of theses by year for recent years.

Year	Number of theses
2012	331
2013	351
2014	386
2015	392
2016	428
2017	448
2018	513
2019	605
2020	633
2021	879
2022	1,212
2023	1,952
2024	2,268
2025	2,236
2026	255

The year field is not used directly by the first retrieval models, but it is useful metadata to display in search results and could later be used as a filter.

3.5 Main disciplines

The dataset is dominated by computer science, but it also contains many theses from applied disciplines. Table 5 reports the most frequent disciplines. The most represented field is *Informatique*, with 3,734 theses. Other frequent disciplines include applied mathematics, physics, applied sciences, neuroscience, economics, management science, language sciences, education sciences and engineering.

Table 5: Most frequent disciplines in the corpus.

Discipline	Number of theses
Informatique	3,734
Mathématiques appliquées	467
Physique	451
Sciences appliquées	436
Neurosciences	403
Mathématiques	323
Sciences et technologies de l'information et de la communication	243
Sciences de gestion	239
Sciences économiques	213
Informatique et applications	203
Informatique, données, IA	197
Sciences de l'information et de la communication	192
Sciences du langage	188
Sciences de l'éducation	157
Informatique et Télécommunications	128

This disciplinary diversity is useful for the retrieval task. A query such as "deep learning for medical imaging" may retrieve theses from computer science, signal processing or medical image analysis. Similarly, a query about natural language processing may retrieve theses from computer science, linguistics, digital humanities or risk management.

3.6 Implications for retrieval

The exploratory analysis leads to three main observations. First, the corpus is large enough to make ranking meaningful, with more than 18,000 theses. Second, the abstract coverage is high, but not complete, which supports the choice of a searchable representation combining titles, English titles and abstracts rather than relying on abstracts only. Third, the corpus is heterogeneous in both metadata and text length. This makes it a realistic retrieval setting, but also makes evaluation more difficult.

These observations motivate the comparison between a lexical baseline and a semantic retrieval method. TF-IDF is expected to perform well when the query and the relevant theses share explicit terms. Transformer embeddings are expected to be more robust when the same concept is expressed with different words.

4 Methods

4.1 TF-IDF retrieval

The first retrieval method is a TF-IDF baseline. The goal is to transform each thesis into a fixed-size vector so that a query and a document can be compared numerically. This follows the classical bag-of-words approach: each dimension of the vector corresponds to a term from the vocabulary, and each coordinate measures the importance of that term in the document.

A simple count representation is not sufficient because very frequent words are not necessarily informative. TF-IDF addresses this issue by combining two quantities. The term frequency measures

how frequent a term is in a document, while the inverse document frequency measures how specific the term is in the whole corpus. A term receives a high weight if it appears frequently in a document but not in too many documents.

For a term t and a document d, the term frequency is defined as:

$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}, \tag{1}$$

where $f_{t,d}$ is the number of occurrences of term t in document d.

The inverse document frequency is defined as:

$$\text{idf}(t, D) = \log \left(\frac{N}{|\{d \in D : t \in d\}|} \right), \tag{2}$$

where N is the number of documents in the corpus and $|\{d \in D : t \in d\}|$ is the number of documents containing term t.

The final TF-IDF weight is then:

$$\text{tfidf}(t, d, D) = \text{tf}(t, d) \times \text{idf}(t, D). \tag{3}$$

In this project, TF-IDF is computed on the final `text` field of each thesis. The vectorizer uses lowercasing, accent stripping, unigrams and bigrams, a minimum document frequency of 2, and a maximum vocabulary size of 50,000 features. The use of bigrams allows the model to capture short expressions such as "deep learning", "medical imaging" or "natural language processing".

Given a query q and a document d, both are transformed into TF-IDF vectors. The relevance score is computed using cosine similarity:

$$\text{cosine}(q, d) = \frac{q \cdot d}{\|q\| \|d\|}. \tag{4}$$

Documents are ranked by decreasing cosine similarity. This method is fast, sparse and interpretable. It is a strong baseline for document retrieval because the terms responsible for a high score can be inspected. However, it remains mostly lexical: it performs well when the query and the document share words, but it may fail when the same concept is expressed with different vocabulary.

4.2 Transformer embedding retrieval

The second retrieval method uses dense semantic embeddings computed with a pretrained multilingual Sentence Transformer. Unlike TF-IDF, which represents documents through sparse lexical features, this approach maps each thesis and each query to a dense vector space where semantic proximity can be measured.

In this project, the model used is `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`. It produces embeddings of dimension 384 and supports multilingual text. This is important because the corpus contains French titles and abstracts, but also English titles and bilingual subject keywords.

No fine-tuning is performed. The model is used as a pretrained encoder because the dataset does not contain supervised query-document relevance labels. Fine-tuning would require annotated examples indicating which theses are relevant for each query.

The embedding workflow is split into two stages. First, document embeddings are computed offline once and saved as a NumPy matrix. For 18,242 theses, this produces a matrix of shape:

$$X_{\text{emb}} \in \mathbb{R}^{18242 \times 384}. \tag{5}$$

Second, at query time, only the query is encoded with the same model. Since document and query embeddings are normalized, cosine similarity can be computed as a dot product:

$$s(q, d_i) = e_q^\top e_{d_i}, \tag{6}$$

where e_q is the query embedding and e_{d_i} is the embedding of thesis i.

This design is computationally efficient. The expensive step, encoding all documents, is done only once. Later searches only require encoding a single query and comparing it with the stored document embeddings.

4.3 Comparison

The comparison focuses on the top-ranked results returned by both methods. For each query, we compare:

- the top retrieved theses;
- the overlap between TF-IDF and embedding results;
- the qualitative relevance of the results;
- cases where lexical retrieval succeeds or fails;
- cases where semantic retrieval brings better results.

5 Experimental setup

The experiments use a fixed set of eight natural language queries. Five queries are written in French, which is the main language of the corpus:

- apprentissage profond pour imagerie médicale;
- traitement automatique des langues pour documents historiques;
- apprentissage automatique pour risque de crédit;
- vision par ordinateur pour diagnostic médical;
- modèles statistiques pour la finance.

Three additional queries are written in English:

- deep learning for medical imaging;
- natural language processing for historical documents;
- machine learning for credit risk.

The English queries are not meant to replace the French evaluation. They are included to study the sensitivity of lexical retrieval to query language. This is relevant because the corpus contains French abstracts and titles, but also English titles and bilingual subject keywords.

For each query, the retrieval model returns the top k most similar theses. The first analysis uses $k = 5$ for manual relevance inspection and $k = 10$ for qualitative exploration. Since the dataset does not provide ground-truth relevance labels, the evaluation is mainly diagnostic and qualitative. Each top-5 result is manually labelled as relevant, partial or irrelevant.

6 Retrieval results

The retrieval methods were evaluated on eight natural language queries. Five queries were written in French, which is the dominant language of the corpus, and three comparable queries were written in English. The English queries were included to test the sensitivity of retrieval to query language, since the indexed documents contain a mixture of French titles, English titles, French abstracts and bilingual keywords.

For each query and each method, the top five retrieved theses were manually inspected and assigned one of three qualitative labels:

- *relevant*: the thesis clearly matches the full query intent;
- *partial*: the thesis matches only part of the query;
- *irrelevant*: the thesis is not related to the query intent.

This evaluation is not a supervised benchmark, but it provides a useful diagnostic of the strengths and weaknesses of both retrieval methods.

Table 6: Manual relevance assessment of TF-IDF top-5 results.

Relevance label	Count	Rate (%)
Relevant	11	27.5
Partial	25	62.5
Irrelevant	4	10.0

6.1 TF-IDF results

Table 6 reports the manual relevance distribution for TF-IDF over 40 inspected results.

The TF-IDF baseline retrieves few completely unrelated documents: only 10.0% of the inspected results were labelled as irrelevant. However, most results are only partially relevant. This is expected for a lexical model: if a document contains some important query terms, it can be ranked highly even when it does not satisfy the full information need.

For example, the query "apprentissage profond pour imagerie médicale" retrieves relevant theses on deep learning and medical imaging. However, other queries show the limitations of lexical matching. For "traitement automatique des langues pour documents historiques", TF-IDF retrieves several theses related to natural language processing, but most are not specifically about historical documents. Similarly, for "machine learning for credit risk", some retrieved results match "machine learning" but are not related to credit risk.

6.2 Embedding results

Table 7 reports the manual relevance distribution for the embedding-based retrieval method over the same 40 inspected results.

Table 7: Manual relevance assessment of embedding top-5 results.

Relevance label	Count	Rate (%)
Relevant	22	55.0
Partial	18	45.0
Irrelevant	0	0.0

The embedding method improves the proportion of fully relevant results from 27.5% to 55.0%. It also removes completely irrelevant results in this manual evaluation. This suggests that dense semantic embeddings better capture the global intent of the query, especially when the relevant documents do not share exactly the same surface words.

The improvement is particularly visible for the query "apprentissage profond pour imagerie médicale". The embedding model retrieves theses on 3D medical imaging, medical image registration, explainability for medical image analysis, computer-aided diagnosis, super-resolution, segmentation and reconstruction. These results are semantically coherent with the query, even when the exact wording differs.

The query "modèles statistiques pour la finance" also shows a strong improvement. TF-IDF retrieved several documents about statistical models outside finance, such as NLP or pharmacovigilance. The embedding model instead retrieves theses on statistical finance, financial applications, high-frequency financial data and financial time series modeling.

6.3 Effect of query language

Table 8 reports the relevance distribution of the embedding method by query language.

The embedding model performs similarly on French and English queries in this small evaluation. The relevant rate is 56.0% for French queries and 53.33% for English queries. This suggests that the multilingual embedding model is less sensitive to query language than a purely lexical method. However, the sample is small, so this result should be interpreted as a qualitative diagnostic rather than a statistically robust conclusion.

Table 8: Manual relevance assessment of embedding results by query language.

Language	Relevant	Partial	Irrelevant	Total
English	8	7	0	15
French	14	11	0	25

6.4 TF-IDF versus embeddings

Table 9 compares the global manual relevance distributions of TF-IDF and embeddings.

Table 9: Comparison of TF-IDF and embedding retrieval over 40 manually inspected top-5 results.

Method	Relevant	Partial	Irrelevant
TF-IDF	11	25	4
Embeddings	22	18	0

The embedding method retrieves twice as many fully relevant results as TF-IDF in this evaluation. It also reduces the number of partial matches and avoids irrelevant documents. This does not mean that TF-IDF is useless. On the contrary, TF-IDF remains a strong, fast and interpretable baseline, especially when the query terms appear explicitly in the indexed fields. However, embeddings are better at ranking documents according to the global semantic intent of the query.

The difference between the methods is also visible through overlap@5. For most queries, the two methods retrieve different top-five documents. The query "apprentissage profond pour imagerie médicale" has an overlap@5 of 2 out of 5, while most other queries have no common document in the top five. This low overlap shows that the two methods use different retrieval signals. TF-IDF ranks documents according to lexical overlap, whereas embeddings rank documents according to semantic proximity.

A low overlap is not necessarily negative. For example, in the medical imaging query, both methods retrieve some relevant theses, but embeddings retrieve a more coherent set of documents related to medical image registration, diagnosis, segmentation and reconstruction. This supports the idea that lexical and semantic retrieval are complementary.

7 Discussion

The experiments show that TF-IDF is a useful lexical baseline, but also highlight its limitations. TF-IDF performs well when the query and the thesis share explicit words or short expressions, such as "apprentissage profond", "imagerie médicale" or "deep learning". It is also fast, sparse and interpretable. However, it often retrieves partial matches when only one part of the query appears in the document.

The embedding-based method improves the quality of the retrieved results in this manual evaluation. It increases the number of fully relevant results from 11 to 22 out of 40 and removes irrelevant results. This suggests that multilingual sentence embeddings are better suited to retrieve documents based on global semantic similarity rather than exact lexical overlap.

The most visible improvement appears in queries combining a method and an application domain. For example, "apprentissage profond pour imagerie médicale" is not only about deep learning, and not only about medical imaging. A good retrieval model must capture the combination of both. TF-IDF sometimes retrieves documents that match only one part of the query. Embeddings more often retrieve theses where the method and the domain are jointly relevant.

The low overlap@5 between TF-IDF and embeddings is also informative. It shows that the two methods retrieve different documents. This can be interpreted in two ways. First, embeddings may correct some lexical false positives from TF-IDF. Second, TF-IDF may still retrieve exact keyword matches that are useful and interpretable. A practical search engine could therefore combine both signals, for example through hybrid retrieval.

Several limitations remain. First, the corpus is not a random sample of all French theses. It was collected through broad keyword queries related to artificial intelligence, machine learning, data science and computer science. Second, abstracts are extracted from HTML pages and may be missing or noisy for some theses. Third, the evaluation is manual and limited to eight queries. A stronger evaluation would require a larger set of queries and independent relevance annotations.

Finally, the embedding model is used without fine-tuning. This is appropriate because no supervised relevance labels are available, but it also means that the model is not specifically adapted to theses.fr data. Future work could explore hybrid retrieval, query expansion, BM25, or supervised fine-tuning if annotated query-document pairs become available.

8 Conclusion

This project builds a reproducible retrieval pipeline for French PhD theses collected from theses.fr. The final corpus contains 18,242 theses, including 16,831 non-empty abstracts extracted from thesis pages. The searchable representation combines titles, English titles, abstracts, disciplines, subjects and institutions.

Two retrieval methods were implemented and compared. The first one is a TF-IDF baseline with cosine similarity. The second one uses pretrained multilingual Sentence Transformer embeddings. The TF-IDF baseline is simple, fast and interpretable, but often retrieves partial lexical matches. The embedding method improves the manual relevance rate from 27.5% to 55.0% on the evaluated top-five results and retrieves no irrelevant document in this small evaluation.

These results suggest that semantic embeddings are useful for searching academic metadata and abstracts, especially when the query combines a method and an application domain or when the relevant vocabulary differs across documents. At the same time, the low overlap between TF-IDF and embeddings indicates that both methods capture different signals. A natural extension would be to build a hybrid search engine combining lexical and semantic retrieval.

References

[1] G. Salton and C. Buckley. Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5):513–523, 1988.

[2] J. Ramos. Using TF-IDF to determine word relevance in document queries. *Proceedings of the First Instructional Conference on Machine Learning*, 2003.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.

[5] N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP-IJCNLP*, 2019.

[6] theses.fr. French PhD thesis search engine. `https://theses.fr`.